Question 2

A1 Sensible choice of equipment and basic idea OK **1**
Source/magnetic field/detector
Inappropriate choice of apparatus cannot score this mark.
Ignore lead or aluminium plates at this stage.

A2 Method of measuring angle of deflection **1**
(e.g. detector at edge of large protractor/lengths & trig ratio used)
Do not allow vague 'use a protractor'.
This mark can be awarded even if the detector has not been specified.

A3 Use Hall probe/search coil/current balance to measure field strength **1**
Allow Helmholtz coils expression if Helmholtz coils used.
Allow a current or voltage measurement as indication of field strength (as $I \propto B$)

B1 Method of removing α radiation or statement that α radiation almost undeflected **1**
Use paper or distance to detector > few cm/air to absorb alpha
Could be shown on the diagram. Do not allow lead/aluminium plate.
Allow α to be shown deflecting in the opposite direction to β on the diagram.

B2 γ -radiation undeflected/deflect beta particles using electric field **1**
Can be shown on diagram. Do not allow 'absorb gamma with lead plate'.

B3 Workable procedure for uniform fields **1**
Measure deflection and field strength; change current in coils and repeat.

C1/2 Any two safety precautions **2**
e.g. use source handling tool
store source in lead lined box when not in use
do not point source at people/do not look directly at source
place lead sheet at 'end of experiment' to absorb unwanted rays

D1/2 Any good/further detail. Examples of creditworthy points might be: **2**
Type of detector (GM tube/film/screen/scintillation counter). N/a cloud chamber/CRO
Repeat readings to allow for randomness of activity
Correct deflection of beta on diagram/left hand rule ideas (diagram or written)
Separation of coils = radius of coils for uniform field
Discussion of count rate (and not just count)
Plane of semiconductor slice is perpendicular to field lines
Calibrate Hall probe
Detail of calibration
Collimation ideas
Allow other valid points. Any two, one mark each.
B1 = B2 = B3 = 0 if lead or aluminium plate is placed in front of the source. Allow thin (less than 1 mm) sheet or foil

10 marks in total.

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